

URYUPINSK, Russian Federation

WMO: 342400

Lat: 50.800N

Lon: 42.000E

Elev: 106

StdP: 100.06

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.1	(c) -20.0	(d) -26.1	(e) 0.4	(f) -22.5	(g) -22.9	(h) 0.5	(i) -19.7	(j) 10.8	(k) -3.4	(l) 10.0	(m) -2.6	(n) 1.8	(o) 330	(p) 0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.6	(c) 33.6	(d) 20.0	(e) 31.5	(f) 19.6	(g) 29.6	(h) 18.8	(i) 21.7	(j) 30.0	(k) 20.9	(l) 28.5	(m) 20.0	(n) 27.3	(o) 4.0	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.1	(b) 14.1	(c) 24.1	(d) 18.3	(e) 13.3	(f) 23.5	(g) 17.5	(h) 12.7	(i) 23.2	(j) 63.7	(k) 30.4	(l) 60.6	(m) 28.6	(n) 57.8	(o) 27.4	(p) 27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 9.1	(c) 8.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 10.0	(b) 9.1	(c) 8.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-6.4	-5.7	-0.9	9.1	15.0	20.1	22.0	20.5	14.2	7.0	-0.2	-4.7
	DBStd	11.55	5.94	6.34	4.94	4.49	4.75	3.80	3.35	3.88	4.72	4.98	6.17	6.78
	HDD10.0	2264	509	438	339	67	11	0	0	0	17	121	306	456
	HDD18.3	4254	768	672	597	276	127	26	5	22	139	353	555	714
	CDD10.0	1377	0	0	0	41	165	303	371	327	142	27	1	0
	CDD18.3	324	0	0	0	0	23	79	118	90	14	1	0	0
	CDH23.3	3404	0	0	0	1	224	768	1258	969	177	7	0	0
	CDH26.7	1272	0	0	0	0	47	261	534	387	43	1	0	0
(14) Wind	WSAvg	3.7	3.7	4.3	4.4	4.2	3.8	3.2	3.0	3.2	3.3	3.4	3.9	3.9
(15) Precipitation	PrecAvg	458	34	28	27	25	40	58	54	33	47	37	37	36
	PrecMax	634	62	62	56	62	87	125	151	85	118	89	76	68
	PrecMin	311	14	9	4	4	11	12	4	3	4	8	2	7
	PrecStd	91	13	13	15	16	18	28	35	21	34	20	21	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	5.7	12.4	23.2	29.8	33.7	36.5	35.1	30.1	23.8	14.6	8.7
		MCWB	5.4	4.5	7.8	12.7	16.9	19.5	21.0	20.5	17.1	15.3	11.9	8.0
	2%	DB	3.1	3.6	9.4	20.9	27.9	31.2	34.3	33.1	27.9	19.8	11.9	5.9
		MCWB	2.3	2.5	5.9	12.0	16.3	20.0	20.4	19.6	16.2	13.9	10.4	5.0
	5%	DB	1.8	2.5	7.0	18.9	25.8	29.7	31.9	30.9	25.5	16.4	9.9	4.0
		MCWB	1.0	1.5	4.6	11.3	16.2	19.4	19.9	18.8	15.7	12.2	8.5	3.1
	10%	DB	0.9	1.6	5.1	17.1	23.5	27.9	29.6	28.6	22.6	14.1	7.7	2.5
		MCWB	0.1	0.7	3.4	10.7	15.3	18.9	19.2	18.2	14.8	10.7	6.5	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	4.8	8.7	14.7	19.7	23.2	22.5	22.2	19.2	16.4	12.5	8.3
		MCDB	5.9	5.6	11.4	19.1	24.6	29.5	32.1	30.1	27.5	21.4	14.4	8.6
	2%	WB	2.7	2.8	6.5	13.5	18.3	21.6	21.7	21.1	17.8	14.9	10.5	5.1
		MCDB	3.1	3.4	8.9	17.9	24.3	27.7	31.3	29.3	23.8	18.9	11.8	5.8
	5%	WB	1.1	1.6	4.9	12.6	17.0	20.4	21.0	20.2	16.5	12.9	8.6	3.2
		MCDB	1.6	2.3	6.7	17.1	22.7	26.8	29.2	28.1	22.8	15.9	9.8	3.8
	10%	WB	0.3	0.8	3.6	11.5	15.9	19.5	20.1	19.2	15.5	10.9	6.7	1.7
		MCDB	0.9	1.5	5.1	16.2	22.2	25.9	28.0	26.2	21.4	13.7	7.8	2.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	6.5	9.8	10.1	10.6	11.6	12.3	10.9	7.5	5.3	4.8
		MCDBR	3.9	4.7	9.3	13.2	13.3	13.8	14.9	15.1	15.7	11.5	6.2	3.8
		MCWBR	3.6	4.2	6.2	6.7	5.4	5.3	4.9	5.0	7.0	6.3	5.2	3.6
	5% WB	MCDBR	3.7	4.5	8.5	10.7	10.7	11.7	13.3	12.8	12.8	10.7	6.4	3.8
		MCWBR	3.5	4.2	6.2	6.2	5.0	5.1	5.1	5.0	6.7	6.3	5.6	3.7
(40) Clear-Sky Solar Irradiance	taub		0.313	0.349	0.375	0.441	0.407	0.397	0.423	0.437	0.391	0.364	0.341	0.315
	taud		2.103	2.001	2.077	2.083	2.282	2.344	2.272	2.214	2.319	2.336	2.263	2.158
	Ebn at Noon		672	742	807	795	849	861	829	793	792	739	647	607
	Edh at Noon		80	117	131	147	126	119	126	128	103	84	70	66
(44) All-Sky Solar Radiation	RadAvg		0.96	1.99	3.34	4.37	5.79	6.17	6.05	5.31	3.63	1.92	0.85	0.56
	RadStd		0.22	0.37	0.53	0.39	0.53	0.53	0.45	0.40	0.42	0.29	0.17	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-137	N/A	N/A

Nomenclature: See separate page